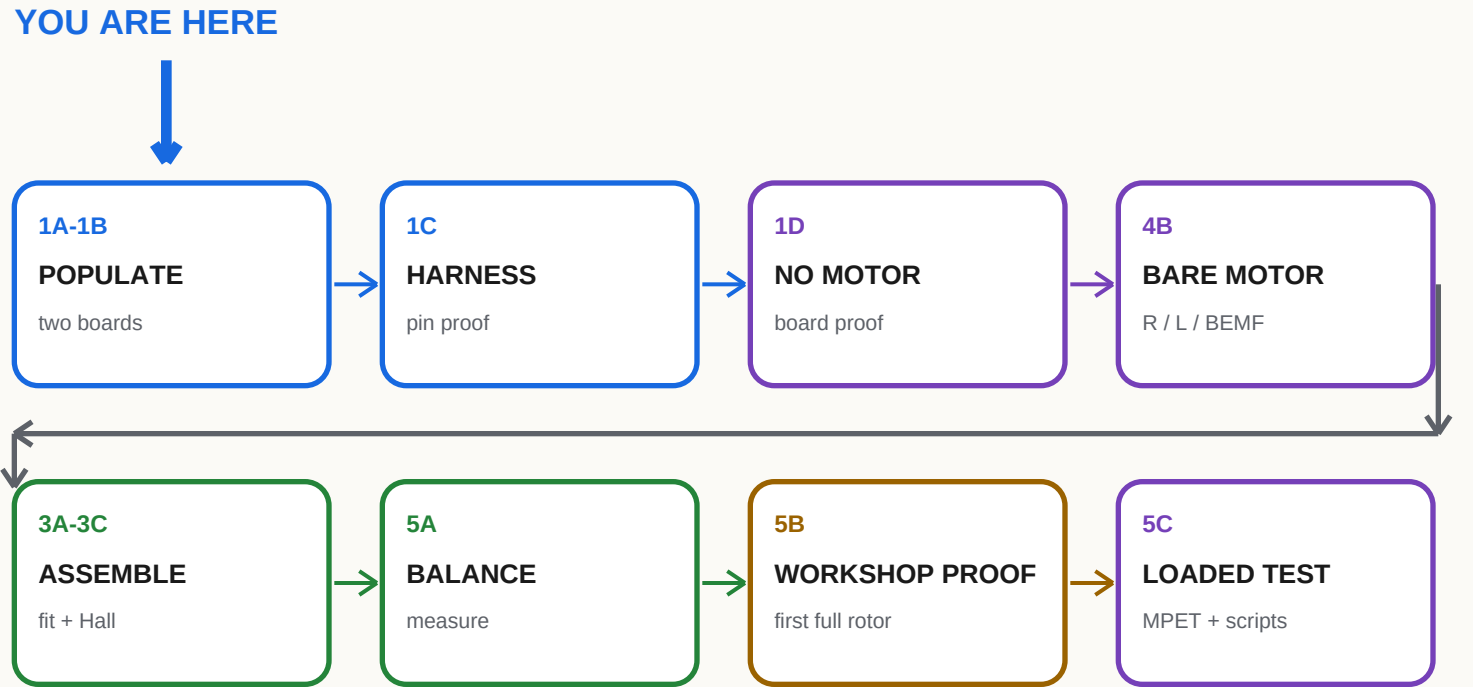




WHAT COUNTS AS DONE

☐ A result is measured and recorded, not just observed.

☐ Connector polarity and pin order are proved end-to-end.

☐ Balance + observed workshop proof precede ceiling loaded work.

☐ A failed gate stops the next powered stage.

☐ After loaded MPET, verify config before start/speed qualification.

☐ Normal safety firmware + long USB J6; cutoff outside sweep.

BOUNDARIES KEPT OUT OF THIS ACTIVE BINDER

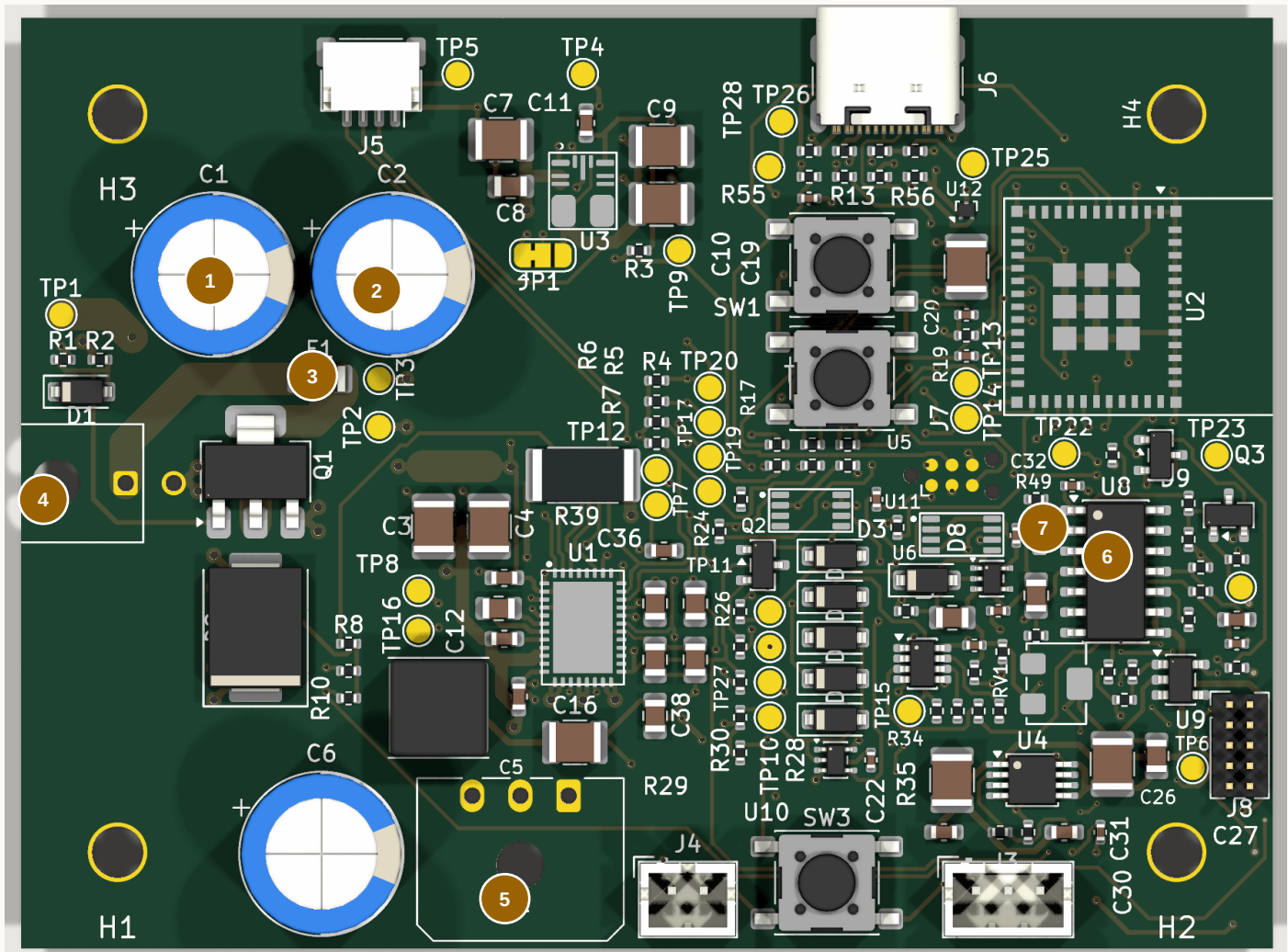
Installed execution remains owner-managed. Sheet 5C is an owner-led reference record; it does not initiate or manage installed work. Deferred scope stays omitted unless explicitly reopened.

STOP / HANDOFF Start at 1A. Carry only the current sheet, board, and required tools to the bench.

1A PCB-01 Population Map

FIELD GUIDE

TOP / COMPONENT SIDE - KiCad target appearance



1 C1
5 J2

2 C2
6 U8

3 F1
7 C34

4 J1

SOLDER ORDER: U8 -> C34 -> J1/J2 -> C1/C2 -> INSPECT -> F1 LAST

1 + 2 C1 / C2

EEU-FR1H471, 470 uF 50 V. Positive lead to board +; negative stripe to pin 2 / PGND. Seat square without crushing bung.

3 F1 - INSTALL LAST

Low-profile tinned copper link across the 1206 pads. Near-zero ohms after soldering. External wall-side 3 A fuse remains mandatory.

4 + 5 CONNECTORS

J1 43045-0200; J2 43650-0300. Larger chisel tip. Tack one electrical pin, square and fully seat housing, then finish. Locating pegs are not soldered.

6 U8 / 7 C34

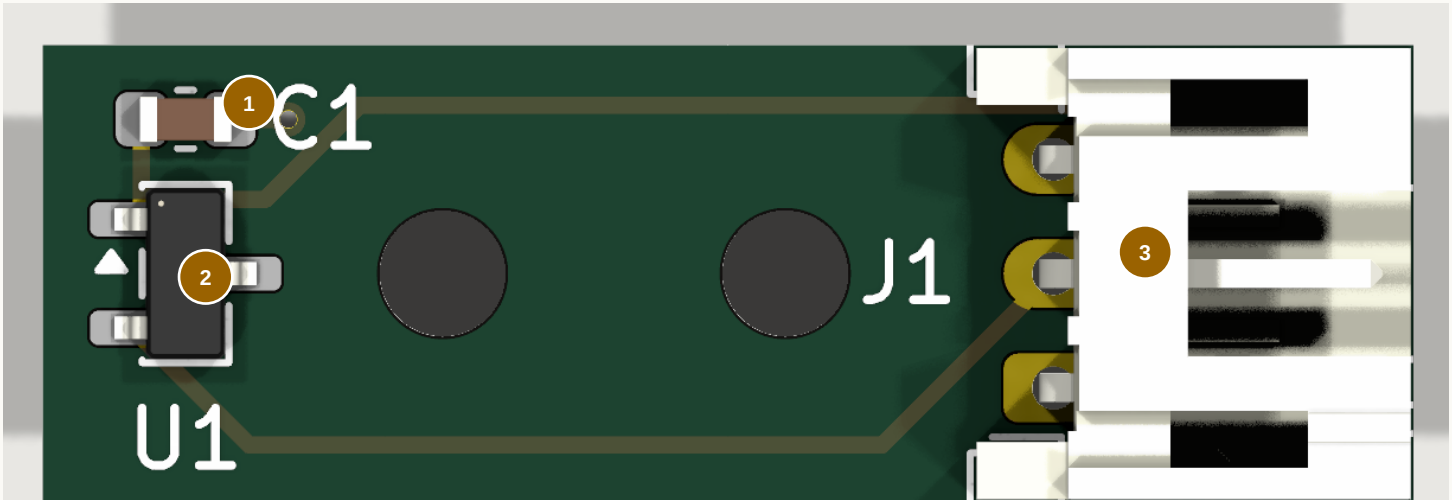
U8 LM2907M/NOPB: notch/dimple to pin-1 marker, tack pin 1 then diagonal pin 8, flux and drag-solder. C34 C1206C104K3GACTU: 100 nF COG 25 V +/-10%, spanning the smaller 0603 site, not shorting it.

PCB-01: USE THE IRON, NOT BROAD HOT AIR. Then clean and dry; inspect under the microscope; verify no hard short from RAW24, VM24, 3V3, or 12V_TACH to ground; photograph the board.

1B PCB-02 Population + Soldering

FIELD GUIDE

TOP / MAGNET-FACING COMPONENT SIDE - KiCad target appearance



J1 / CABLE EXITS RIGHT →

1 C1

C0603C104K5RACTU, 100 nF 50 V X7R, 0603. Pin 1 / 3V3 is left; pin 2 / AGND is right.

2 U1

DRV5033FAQDBZR, SOT-23. Match package index to board triangle. Pin 1 upper-left = 3V3; pin 2 lower-left = HALL_TACH; pin 3 right = AGND.

3 J1

S3B-PH-K-S side-entry JST-PH. Top to bottom in this view: 3 AGND, 2 HALL_TACH, 1 3V3. Iron-solder last.

IRON METHOD: THIN SOLDER WIRE + FLUX

Flux + tin one pad

Place under microscope

Reheat to tack

Solder remaining pads

Inspect + wick bridges

Iron-solder J1

C1 - ONE-PAD TACK

Tin one pad lightly. Reheat it while placing C1, align the body, then solder the other end. Refresh the tack only if needed.

U1 - THREE LEADS

Flux, tack pin 3, correct alignment, then solder pins 1 and 2. Reflow the tack last if it needs a cleaner fillet.

J1 - LAST

Tack one pin, reheat while seating the body square and flush, finish the other pins, then revisit the tack.

FINAL INSPECTION

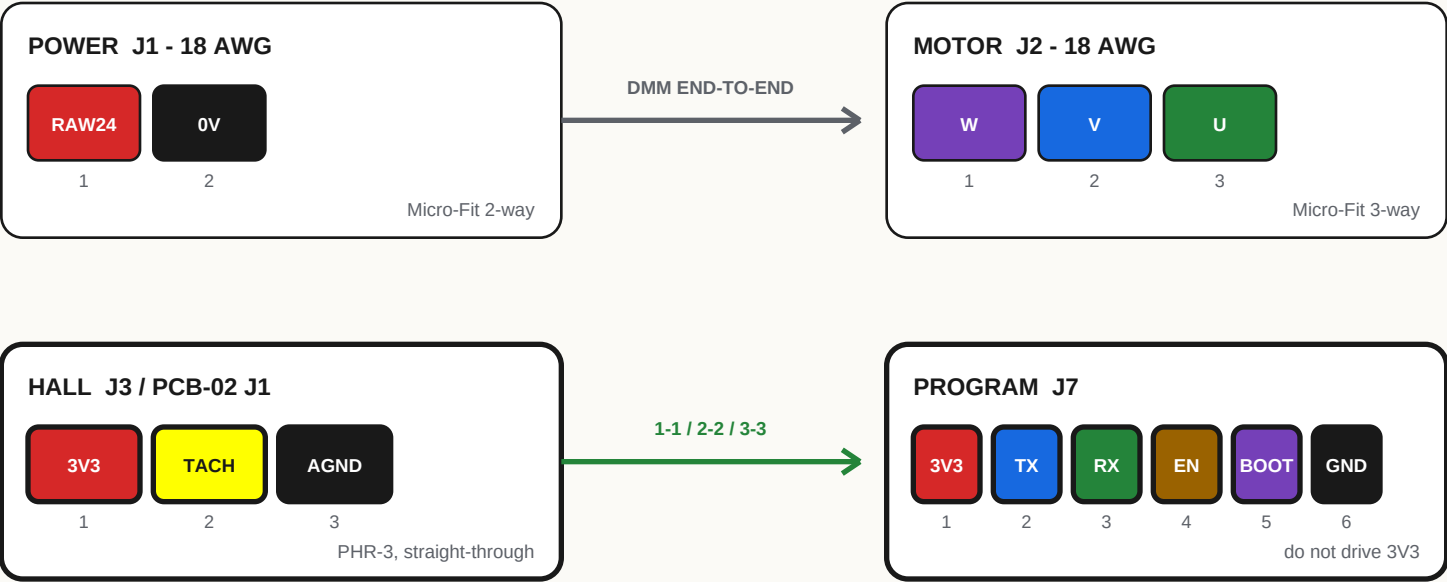
- ☐ All parts on magnet-facing side; sensor marked face will face magnet. Do not use magnet polarity as the orientation cue.
- ☐ Flux cleaned using its specified method; board completely dry.
- ☐ No short: 3V3 to HALL_TACH, 3V3 to AGND, or HALL_TACH to AGND.
- ☐ Microscope: wet fillets, no lifted lead, whisker, ball, or disturbed part; photograph board.

STOP / HANDOFF

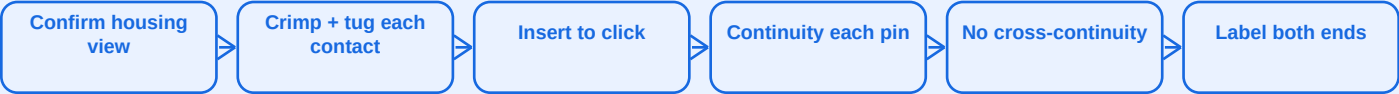
PCB-02 passes magnified inspection and unpowered three-net short checks. Do not power yet.

1C Harness Build + Pin Proof

LOOK INTO THE MATING FACE; USE MOLDED PIN-1 MARKS, NOT WIRE COLOR MEMORY



EACH HARNESS: BUILD -> PROVE -> LABEL



- ☐ Power: +24 V and 0 V preserved end-to-end.
- ☐ Motor: label positions W / V / U even if flying leads are not yet identified.
- ☐ Hall: exactly 1-to-1, 2-to-2, 3-to-3.
- ☐ Record length, gauge, wire colors, contact orientation, and result before sleeving.

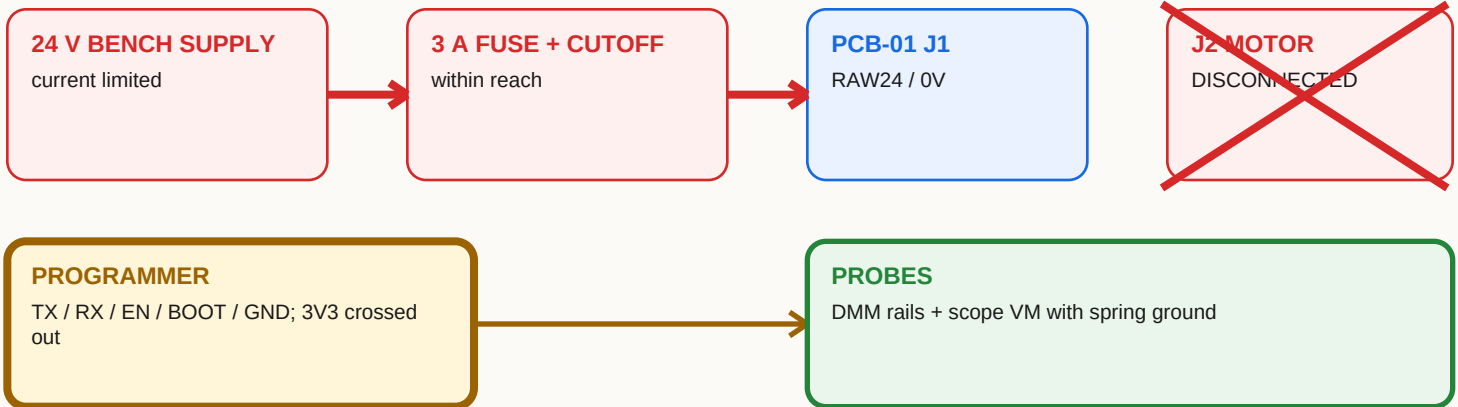
J7 PIN 1 IS BOARD 3V3, NOT A POWER INPUT. Never let the programmer drive it while 24 V is connected.

STOP / HANDOFF Every harness is labeled, pull-tested, and signed off for pin continuity and no cross-continuity.

1D PCB-01 First Power, No Motor

FIELD GUIDE

WIRE THIS FIRST; PLACE PROBES BEFORE POWER



NUMBERED TEST FLOW

1 RAILS

Brief power: 24 V, 3.3 V, AVDD, DVDD, DRVOFF; record current and heat.

2 HARDWARE CUTS

Force PGOOD, WDO, manual clear, OVERSPEED_N low one at a time -> DRVOFF rises without ESP help.

3 WATCHDOG

Static WDI: timeout 1.44-1.76 s, WDO pulse about 200 ms. 2 Hz falling edges prevent timeout.

4 STARTUP

Cold + slow-ramp power-up 20 times: U6 /PRE releases only after TACH_PGOOD_N.

5 RECOVERY

Remove/restore 24 V: disabled ≥ 10 s; requires new command plus explicit arm.

6 TACH

Bridge disabled. Inject 0-3.3 V, 1-3% duty, settle ≥ 30 s: reset ~ 3.00 Hz, trip ~ 3.33 Hz.

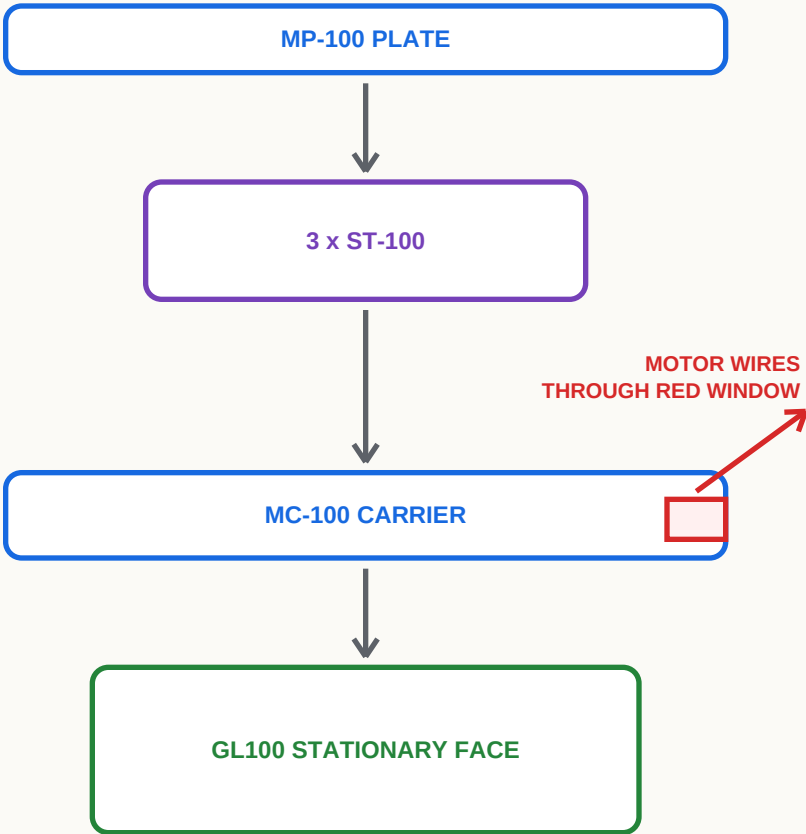
PASS LIMITS: VM ≤ 35 V target; 40 V rejects. Persistent tach lock clears only on a genuine low-voltage power cycle. Any automatic re-arm blocks release.

STOP / HANDOFF

All listed board/safety tests have recorded passes. Do not connect the motor after any failure.

3A Stationary Stack Dry-Fit

EXPLODED SIDE VIEW - NON-POWERED



MP-100 -> ST-100
3 x M6 x 16 flat-head from plate ceiling face; heads flush or below.

ST-100 -> MC-100
3 x M6 x 20 A4-80 + Nord-Lock pairs. Hand-start, square seating.

MC-100 -> GL100
4 x M4 x 12 A4-80 into the 60 mm stationary pattern. The 50 mm face rotates.

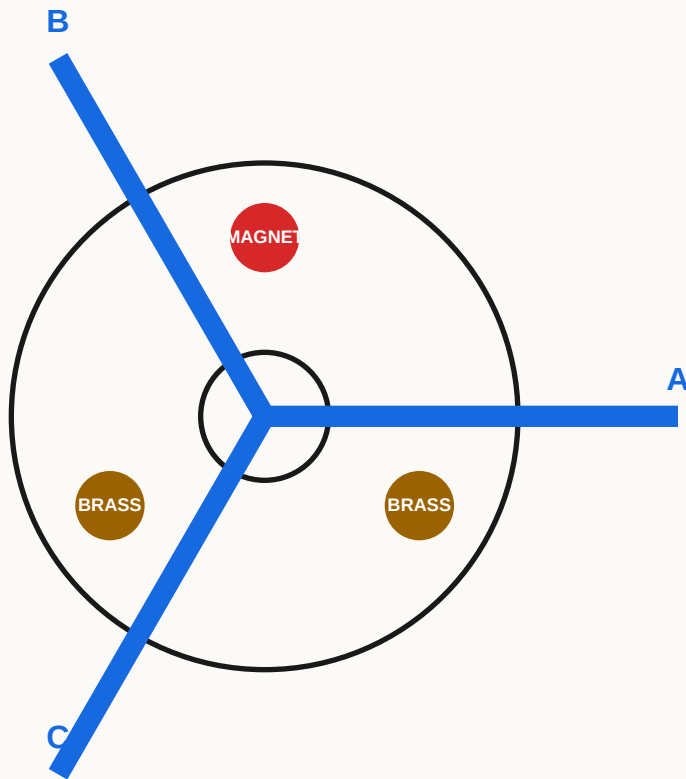
THREAD DEPTH
Target about 5.5 mm engagement; 6.0 mm motor-thread maximum. A bottoming screw is felt at hand torque: stop and correct.

RELEASE CHECK

- | | |
|---|---|
| ☐ All fasteners hand-start; no rocking or forced alignment. | ☐ Motor wire is unpinched and clear of sharp edges. |
| ☐ Owner-selected torque applied consistently; witness marks present. | ☐ Threadlocker only where specified; no screw bottoms. |
| ☐ Stationary 60 mm and rotating 50 mm faces confirmed. | ☐ Stack remains non-powered until later gates pass. |

STOP / HANDOFF Dry stack sits square without force, rocking, wire pinch, or bottoming screws.

3B Rotor + Tach Inserts



HUB TO MOTOR

4 x M4 x 10 A4 flat-head from underside into 50 mm rotating pattern; heads 0.1-0.2 mm subflush. Pilot enters bore by hand only.

EACH BLADE

BP-100 v3 locating pins + captured M5 nuts; 4 x M5 x 22 per blade. Torque consistently and witness-mark.

THREE INSERT STATIONS

Epoxy retains one magnet + two brass slugs. Match complete insert-plus-epoxy station masses within 0.01 g after cure.

MASS + FINAL RECORD

MAGNET _____ g BRASS B _____ g BRASS C _____ g MAX SPREAD _____ g

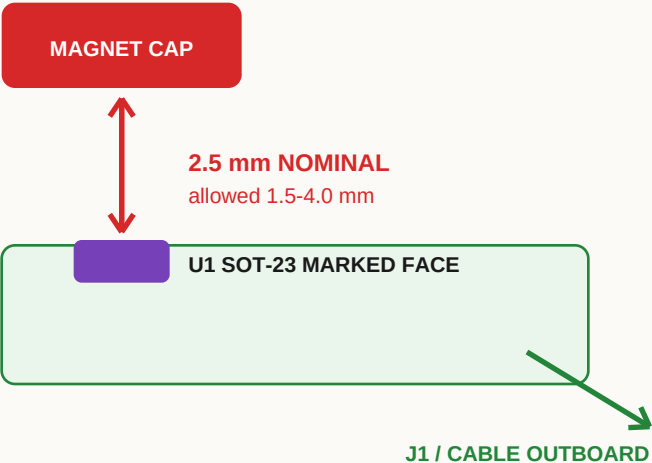
- | | |
|---|---|
| ☐ Pilot hand-fit; all four hub heads subflush. | ☐ All three blade stations have four correct bolts and captured nuts. |
| ☐ Epoxy fully cured; inserts cannot move. | ☐ Hand-rotate several revolutions: no rub, click, or changing clearance. |
| ☐ Critical fasteners torqued and witness-marked. | ☐ Photograph rotor before balance measurements. |

Do not use cad/BP-100.step for current fit; it is stale v2 geometry.

STOP / HANDOFF

Complete rotor turns freely with correct retention; continue to 5A before powered motion.

SECTION THROUGH MAGNET AND SENSOR



COMMON CENTERLINE

Magnet and sensor element at radius 76.0 +/-0.5 mm.

PCB-02 MOUNTING

M2 holes on 6 mm pitch. At H1 use bare pan head or washer <=4.5 mm OD. Components face magnet.

BR-100

Strain-relieve Hall cable. Validate bracket around physical motor, board, magnet, and real gap.

PCB-01 BRACKET ENVELOPE

110 x 80 x 25 mm SERVICE ENVELOPE

Four 6-8 mm M3 standoffs. Keep isolated mounting holes clear of circuit ground. Independent PCB retention and cable clamps. Preserve connector and cable-bend keepouts.

HAND-ROTATION RELEASE

- ☐ One clean pulse/revolution with magnet.
- ☐ All cables and connector bodies clear rotor.
- ☐ No false pulse with magnet removed.
- ☐ Gap remains in range through full hand rotation.

STOP / HANDOFF

Hall pulse is clean; retained PCB and clamped cables clear all moving parts.

BUILD / FLASH / TALK

```
cargo build --manifest-path firmware/Cargo.toml
```

```
esptool flash --port /dev/cu.usbmodem2101 --non-interactive  
firmware/app/target/riscv32imac-unknown-none-elf/debug/stillair
```

```
firmware/target/debug/stillair --port /dev/cu.usbmodem2101 state
```

BARE DEV-BOARD INPUT JUMPERS

GPIO22 -> 3V3

PGOOD good

GPIO21 -> 3V3

nFAULT idle high

GPIO14 -> GND

ALARM idle low

SafeBoot

Starting

15 s, no FG

NoRotation fault

EXPECTED ON A BARE BOARD; IT DOES NOT GENERATE FG EDGES

SESSION RULES

ONE READER

Stop raw-log capture before esptool or CLI opens the port.

USE SCRIPT

Multi-step simulator work must stay in one session; separate invocations reset it.

RELEASE GATE

config check must report config=verified. ok=true plus unverified is not a pass.

wait <state> --for <seconds> | wait speed <rpm> --within <rpm> --for <seconds>

stream <hz> --for <seconds> for CSV | config capture after complete measured configuration

Firmware refuses raw register/config writes unless stopped. Use controlled mpet run, not live raw commands. Simulator passes validate the harness, not the motor.

STOP / HANDOFF

Firmware flashes; one persistent CLI session communicates and preserves scripted state.

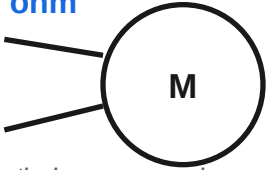
4B Desk: Bare GL100 Measurements

HOLD: IMAGE UNVERIFIED

NO BLADES x Motor secured; manual cutoff ready; board/safety tests already passed

1 PHASE RESISTANCE

ohm



method: _____ ohm

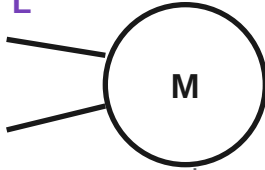
U-V: _____

V-W: _____

W-U: _____

2 PHASE INDUCTANCE

L



frequency: _____ L

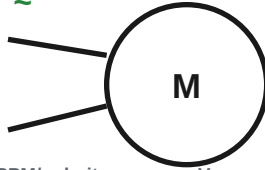
U-V: _____

V-W: _____

W-U: _____

3 MANUAL-SPIN BEMF

~



RPM/polarity: _____ Vpp

U-V: _____

V-W: _____

W-U: _____

Repeat across phase pairs; record connection and polarity convention. Confirm 20 pole pairs.

GREEN LANE - DO NOW

- ☐ Measure R and L independently with method recorded.
- ☐ Manually spin only; scope line-to-line BEMF and convention.
- ☐ Use script 03 at the provisional minimum only; keep current/speed caps.
- ☐ Keep loose hub hardware and blades off the motor.

LOADED-STAGE HANDOFF

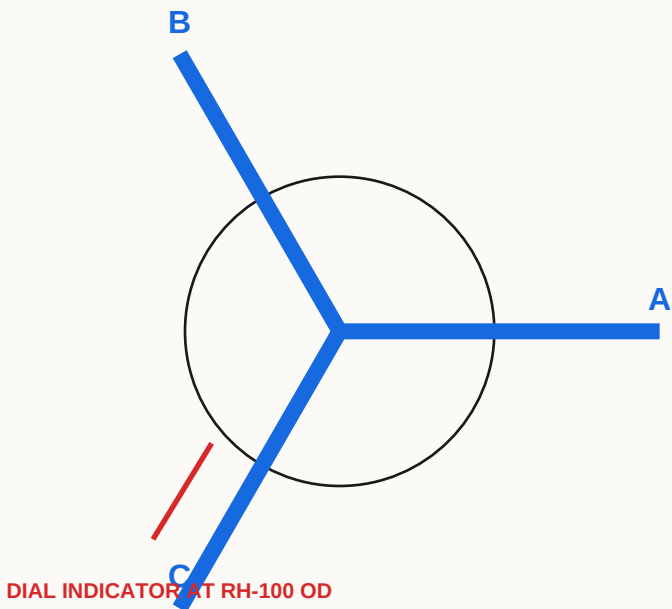
- ☐ No unloaded MPET; script 02 requires the representative loaded rotor.
- ☐ Golden image still needs measured D-generation fields and review.
- ☐ config apply is stopped-only and confirms commit + readback.
- ☐ After power cycle require config check: verified, not merely ok=true.

No live raw register writes. For any later powered work, VM <=35 V target; 40 V rejects.

STOP / HANDOFF R / L / BEMF records are complete. Loaded MPET waits for the rotor, not for more software.

5A Rotor Balance + Runout Record

FIELD GUIDE



HUB RUNOUT
≤0.10 mm TIR

BLADE TIP HEIGHT
all three in one indexed setup; spread ≤0.5 mm

FIRST MOMENT
each blade; spread ≤0.5%

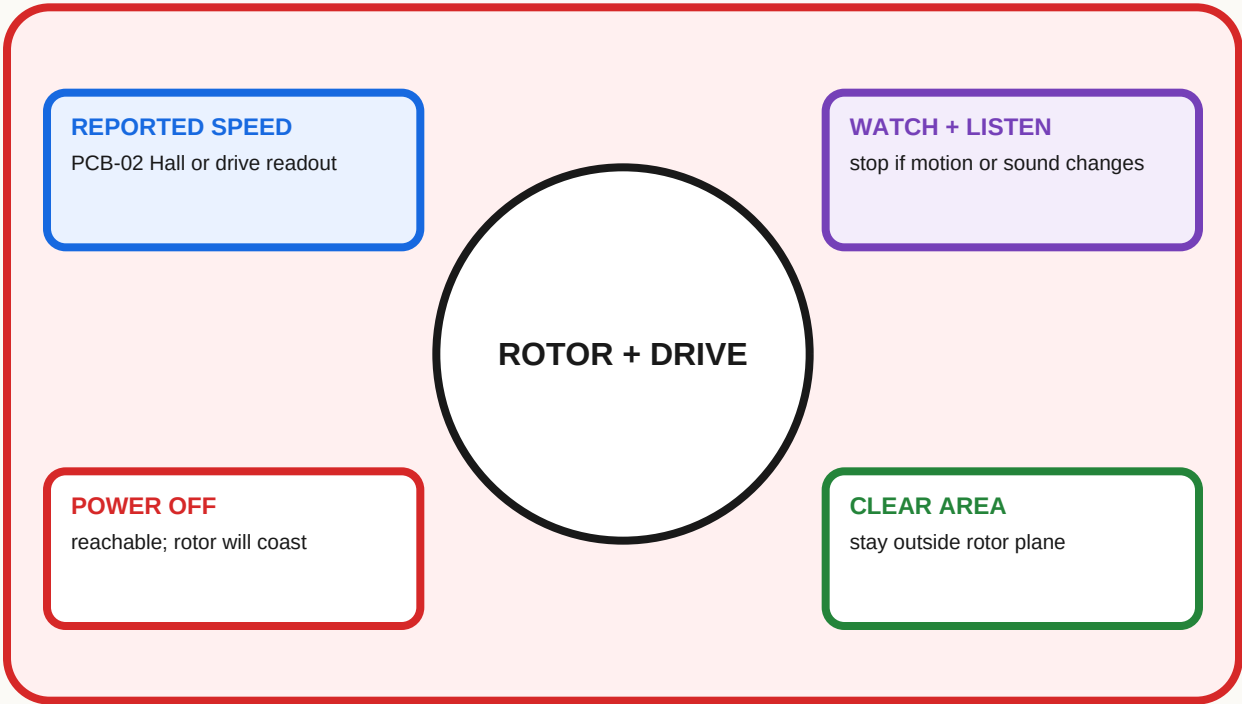
A / B / C MEASUREMENT TABLE

STATION	MASS (g)	FIRST MOMENT	TIP (mm)	CORRECTION
A				
B				
C				

METHOD CHECK

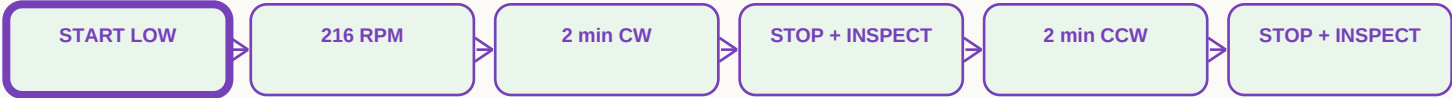
- ☐ 0.01 mm indicator + 0.01 g calibrated scale.
- ☐ First powered check starts low; stop for wobble, vibration, rubbing, or unusual sound.
- ☐ Correct only by documented slug method; remeasure.
- ☐ Hand-clearance pass and witness-mark photo saved.

STOP / HANDOFF MEC-05 passes numerically; do not advance a rotor that only looks balanced.



GL100 PHASES DISCONNECTED FROM PCB-01 | NO SAFETY BYPASS

PCB-02 MAY REPORT HALL SPEED; ANALOG TRIP MAY LATCH BUT WILL NOT STOP THE EXTERNAL DRIVE



270 RPM = CALCULATION ONLY x NEVER DYNAMICALLY TEST

BEFORE, DURING, AND RECORD

- ☐ Setup secured; rotor plane and nearby area clear.
- ☐ Start low; advance only while smooth and quiet.
- ☐ Balance, runout, clearance, witness marks recorded.
- ☐ Speed source responds and is recorded.
- ☐ Watch continuously; cutoff remains reachable.
- ☐ Inspect after each direction; stop on any change.

Speed source: _____ Setup: _____

CW actual: _____ RPM result: _____ CCW actual: _____ RPM result: _____

Notes: _____

Date / initials: _____

STOP / HANDOFF

Both directions pass with no abnormal motion, sound, contact, loosening, or damage.

5C Loaded Commissioning Script Card

HARDWARE VALUES PENDING

FINAL TEST CONNECTION

LAPTOP + CLI

LONG USB J6

PCB-01

MOTOR + ROTOR

NORMAL SAFETY FIRMWARE | 24 V VIA REACHABLE CUTOFF | CABLES OUTSIDE SWEEP

CONTROLLED MPET + GOLDEN IMAGE

- ☐ Run scripts/02-mpet-and-capture.txt only with the representative loaded rotor.
- ☐ Put only reviewed D-generation values into the golden IMAGE.
- ☐ Power-cycle after apply; config check must report verified.
- ☐ Review MTR_PARAMS, CURRENT_PI, and SPEED_PI against independent R / L / BEMF.
- ☐ While stopped, config apply performs one commit, waits 750 ms, polls self-clear, then verifies readback.
- ☐ If MPET faults or times out, firmware aborts it and revokes drive permission.

MPET result / image revision: _____

RUN THE NUMBERED FILES

02	scripts/02-mpet-and-capture.txt	loaded MPET + config capture
04	scripts/04-loaded-speed-ladder.txt	35 / 60 / 90 / 120 / 150 / 170 RPM ladder
05	scripts/05-direction-check.txt	low-speed direction check through verified stop
06	scripts/06-observed-run.txt	30-minute continuously observed run

WATCHED STOP RULES

- ☐ Keep the ordinary plug/cutoff reachable and watch continuously.
- ☐ Stop for visible wobble, increasing vibration, rubbing, or unusual sound.
- ☐ Use reported FG diagnostics and your own observation; no external tachometer is required.
- ☐ No permissive firmware or safety bypass. A stopped run may be repeated after inspection.

STOP / HANDOFF

Loaded MPET, verified config, speed ladder, direction check, and observed run have recorded results.